

SUPPLY-SIDE RESPONSES TO TARGETED VOUCHERS*

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Abstract

We examine schools' responses to a recent targeted voucher reform in Chile. Private schools' responses to the reform are quicker and larger than public schools' responses; however, schools in both sectors react to the policy along various margins. Private schools enter and exit the market more often. Public and private schools improve educational inputs related to school infrastructure and the teaching staff. Interestingly, private schools' fee responses are such that low income students see an increase in the average fee they face in the market, and higher income students see a decrease in the average fee they face.

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1 Introduction

The literature on school vouchers often focuses on the demand side of education markets: students and their families.¹ Studies ask what determines parental school choice and how attending a private school affects student achievement relative to attending a public one. These are important policy questions, but voucher policies also change the incentives and regulations faced by schools. Schools' responses can, in turn, shape the effects of these policies on student outcomes (Neilson, 2020; Dinerstein and Smith, 2021). For instance, a voucher program that induces schools to improve educational quality can raise test scores even without changes in student sorting. Understanding these supply-side responses is therefore of first-order importance for policy design and for explaining the overall effects of voucher programs.

In this paper, we study schools' responses to Chile's 2008 targeted voucher reform. The reform considerably increased government subsidies and introduced new rules that changed schools' incentives. We examine responses along both the extensive margin, through entry and exit, and the intensive margin, through educational inputs and top-up fees.

In 1981, Chile introduced a universal voucher that provided public and subsidized private schools with a flat subsidy for each student enrolled. We refer to these subsidized private schools as private-voucher schools. In addition to the universal voucher, private-voucher schools could charge top-up fees to families. The 2008 reform supplemented the universal voucher with a targeted subsidy for economically disadvantaged students. Schools could choose whether to participate in the new program. Those that joined received additional funds for each eligible low-income student they enrolled and were required to waive top-up fees for those students. The targeted voucher initially amounted to about 60% of the universal voucher, representing a considerable budgetary effort by the government.

We identify the reform's effects by exploiting differences in exposure across local education markets, which we define as municipalities. Our measure of treatment intensity is the share of eligible low-income students in a municipality before the reform. This share captures the market's ex-ante potential to receive additional funds. A municipality with no eligible students cannot receive the targeted subsidy, whereas one in which all students are eligible has the greatest potential exposure. We fix this share in 2007, the year before the reform, to avoid endogeneity from residential migration in response to the program. Our event study compares outcomes before and after the reform across markets with different treatment intensities (Card, 1992; Duflo, 2001; Lucas and Mbiti, 2012). The coefficients measure the effect of full versus null treatment: a municipality in which all students are eligible compared with one in which none are eligible. We estimate these effects separately for public and private-voucher schools. Because treatment is defined at the mar-

¹See the extensive review in Eppele et al. (2017).

ket level, the estimates include the responses of participating and non-participating schools, as well as equilibrium adjustments in the market.

We find that both public and private-voucher schools respond to the reform along several margins. Private-voucher schools respond more quickly and more strongly, consistent with the arguments in Pritchett (2013). Public schools take longer to adjust, but many of their responses become statistically significant by the end of our study period (Hoxby, 2003b). On the extensive margin, the reform increases both entry and exit among private-voucher schools. Public schools are unresponsive along this margin.

Both public and private-voucher schools also adjust their educational inputs. These adjustments generally move in the direction of improving educational quality, through both infrastructure and teachers' working conditions. Both sectors make costly adjustments to infrastructure-related inputs, reducing class size and the share of multigrade classes, which combine students from different grades. The responses in staffing differ across sectors: private-voucher schools reduce their pupil-teacher ratios, while public schools increase them. Schools maintain or improve the working conditions we observe for their teachers. In both sectors, the share of teachers on long-term contracts remains unchanged, while teaching hours per teacher decline.

The share of specialized teachers follows a different pattern.² Private-voucher schools initially reduce this share, but reverse the response so that it is unaffected by the reform by the end of the study period. Public schools follow a similar pattern with a lag, and we do not observe a complete reversal within our period of analysis.

Possibly the most interesting result concerns private-voucher schools' top-up fees. Before the reform, these schools could charge a single fee to all students. Non-participating schools retained this pricing rule. Participating schools, however, had to charge zero fees to eligible low-income students while remaining able to charge positive fees to higher-income students. The reform thus introduced a form of third-degree price discrimination at participating schools. We therefore distinguish between the fees faced by eligible low-income students and those faced by higher-income students.

Relative to null treatment, full treatment increases the average top-up fee faced by eligible low-income students and reduces the average faced by higher-income students. These averages give each school equal weight and describe the fees available to each eligibility group across schools in the market. The increase for eligible students is particularly surprising given the requirement that participating schools charge them zero fees. We also document how fee distributions and average fees evolve among schools grouped by their participation status in 2011.

This paper contributes to the literature along two main fronts. First, it adds to the evidence on the effects of voucher programs. Epplé et al. (2017) review studies of the effects on voucher recipi-

²A specialized teacher is a teacher with extra training in one or more subjects.

ents, student sorting across schools, and competitive pressure on public schools. They also review evidence on overall educational performance and the political economy of voucher programs. Our contribution is to document school responses at the market level, including equilibrium adjustments. These responses are a key mechanism through which vouchers affect student achievement. For instance, Abdulkadiroglu et al. (2018) argue that the participation of low-quality private schools was a major driver of the Louisiana Scholarship Program’s negative effects on student test scores.

Our findings for public schools also speak to the literature on their productivity, much of which focuses on the United States (Hoxby, 2003a; Figlio and Rouse, 2006; Chakrabarti, 2008, 2013a,b; Figlio and Hart, 2014).³

Second, our paper contributes to the literature on the industrial organization of education markets. Studies in this literature typically develop and estimate structural models of demand, supply, or both. They use these models to study parental preferences for schools, schools’ strategies under imperfect competition, and the equilibrium consequences of actual and counterfactual policies.⁴ We are motivated by similar questions about competition and regulation. We address these questions using a clean reduced-form empirical design, which allows us to examine a broader set of outcomes than structural models often do.⁵

The remainder of the paper is organized as follows. Section 2 describes Chile’s education system and the targeted voucher program. Section 3 describes our data. Section 4 presents the identification strategy and empirical results. Section 5 concludes.

2 The Chilean School System and the Targeted Voucher Program

During the period we study, Chile’s school system comprised three types of schools, distinguished by their management and financing: public schools, private-voucher schools, and private-non-voucher schools. Public and private-voucher schools received per-student voucher subsidies paid by the government directly to the schools. Private-voucher schools could also charge top-up fees to families. Private-non-voucher schools were financed entirely by fees and served the country’s richest families. In 2021, 38% of primary school students attended public schools, 53% attended

³See also Hsieh and Urquiola (2006), who show that public schools’ performance worsened in markets with substantial private-sector competition after the introduction of Chile’s universal voucher.

⁴See, for example, Gazmuri (2015), Ferreyra and Kosenok (2018), Allende (2019), Allende et al. (2019), Singleton (2019), Dinerstein et al. (2020), Neilson (2020), Sánchez (2026), and Dinerstein and Smith (2021).

⁵Dinerstein et al. (2020) and Dinerstein and Smith (2021) also examine schools’ responses to policies using reduced-form strategies.

private-voucher schools, and 9% attended private-non-voucher schools.

Chile introduced the universal voucher in 1981. Under this system, every student was entitled to a flat subsidy that could cover at least part of tuition at any public or private-voucher school of her choice. In 2008, the government supplemented this subsidy with a targeted voucher for economically disadvantaged students. The *Ley de Subvención Escolar Preferencial (SEP)* was enacted in February 2008 and implemented that same academic year.⁶

Participation in the targeted voucher program was voluntary and open only to public and private-voucher schools. Participating schools received an additional subsidy for each eligible low-income student they enrolled and were required to charge those students no top-up fees. The program also included a *Subvención por Concentración*, or concentration subsidy. The amount of this per-student subsidy depended on the share of eligible low-income students enrolled in the school. Participating schools were also required to set short- and long-term learning goals, expressed in terms of test scores, which the government evaluated every four years.

The eligibility criteria were designed to cover households in the lowest 40% of the income distribution. Although eligibility depended on several criteria, two were particularly important. One was whether the household fell within the lowest 33% of the socioeconomic index used in the *Ficha de Protección Social*, which guided the government’s allocation of social assistance. The other was participation in *Chile Solidario*, a social program serving vulnerable families.⁷

By 2011, approximately 48% of students in Chile were eligible for the targeted voucher. Of these students, 88% attended participating schools, representing 42% of all students in the country. The program thus directly reached almost half of the student population.

Table 1 presents monthly per-student subsidies for grades 1–4 over the period 2004–2015, distinguishing the universal voucher, the targeted voucher, and the concentration subsidy. The amounts are expressed in real US dollars and correspond to schools operating a full school day. The universal voucher is set to reflect the unit cost of educating a student. Its value increases overall from approximately \$63 in 2004 to \$93 in 2015. The targeted voucher is set to reflect the additional unit cost of educating a low-income student. It begins at approximately \$37 in 2008 and reaches \$57 in 2015. The concentration subsidy is considerably smaller and remains fairly stable as a share of the universal voucher. It amounts to roughly 4–11% of the universal voucher, depending on the share of eligible students enrolled in the school.

⁶Chile’s academic year runs from March to December.

⁷See Neilson (2013, 2020) and Aguirre (2020) for further details on student eligibility.

Table 1: Monthly Per-Student Voucher Subsidies for Grades 1–4

Category	Subsidy (US\$)											
	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015
Universal voucher	62.66	63.07	64.92	63.82	62.40	80.86	82.48	82.71	84.77	91.50	90.93	92.62
Targeted voucher	–	–	–	–	36.81	42.86	43.71	43.84	44.93	55.93	55.59	56.62
Concentration subsidy:												
15–30%	–	–	–	–	2.66	3.00	3.06	3.06	3.14	3.89	3.87	3.94
30–45%	–	–	–	–	4.56	5.14	5.24	5.26	5.39	6.67	6.62	6.75
45–60%	–	–	–	–	6.08	6.85	6.99	7.01	7.18	8.88	8.82	8.99
<i>more than 60%</i>	–	–	–	–	6.84	7.71	7.86	7.89	8.08	9.97	9.91	10.09

Notes: All amounts are expressed in real US dollars, converted from Chilean pesos at the May 16, 2016 exchange rate of Ch\$686.52 per US dollar. Universal voucher amounts correspond to schools operating a full school day. The percentage ranges indicate the share of eligible low-income students enrolled in the school.

The evidence on the program’s effects on student outcomes is encouraging. There is wide consensus that the program raises test scores (Correa et al., 2014; Sánchez, 2019) and narrows income-based achievement gaps (Navarro-Palau, 2017; Feigenberg et al., 2019; Neilson, 2020). These findings indicate that the most vulnerable students benefit the most from the program.⁸ Our examination of schools’ responses follows naturally from this evidence, as a means of understanding the mechanisms behind the gains in student achievement.⁹

3 Data

We combine administrative data from Chile’s Ministry of Education covering the years 2004–2015. These data include the censuses of students, schools, and teachers.¹⁰ We use them to construct two twelve-year unbalanced panels: one at the school level and one at the municipality level. The underlying data cover the universe of Chilean schools offering at least one grade from 1 to 8, which constitute primary education.

Table 2 reports the number of schools by type and participation in the targeted voucher program over the period 2004–2015. Panel A shows the total number of schools and its composition by school type. The total falls from 8,908 in 2004 to 8,425 in 2015, a modest decline that may

⁸An exception is Aguirre (2020), who uses a regression discontinuity design based on the student eligibility index and finds no statistically significant effects on test scores. These findings need not contradict the positive effects found in studies using difference-in-differences designs. Her estimates concern the effect of eligibility for the targeted voucher when the program is already in place, whereas those studies estimate the effects of the reform. Moreover, null effects near the eligibility cutoff do not imply that effects are absent elsewhere in the eligibility distribution.

⁹Feigenberg et al. (2019) examine the reform’s effects on class size and selected teacher characteristics. Their results are largely consistent with our findings on educational inputs.

¹⁰Appendix A describes each data set in detail.

reflect demographic changes and shifts in family-size preferences in Chile. Public and private-voucher schools account for about 94% of these schools, illustrating the targeted voucher program’s substantial ex-ante potential reach.

Public schools remain the most numerous, but their number falls from 5,426 in 2004 to 4,613 in 2015. Over the same period, the number of private-voucher schools rises from 2,928 to 3,386. These contrasting trends are consistent with estimated models of families’ preferences for school types in Chile.¹¹ Private-non-voucher schools account for a small share of the total, and their number falls from 554 in 2004 to 426 in 2015.

Panel B documents the program’s ex-post reach. Almost all public schools join immediately: 97% participate in 2008, the year the program is introduced. Their participation remains almost universal throughout the period, reaching 99% in 2015. Among private-voucher schools, participation begins at 47% in 2008 and rises rapidly to 75% in 2015. These figures show that the program achieves broad participation in both sectors.

Table 2: Schools by Type and Participation in the Targeted Voucher Program

	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015
<i>A. All schools</i>	8,908	8,924	8,871	8,839	8,827	8,848	8,804	8,699	8,674	8,573	8,491	8,425
Public	5,426	5,397	5,267	5,192	5,126	5,080	5,004	4,882	4,826	4,740	4,660	4,613
Private-voucher	2,928	3,068	3,159	3,202	3,259	3,336	3,365	3,390	3,425	3,414	3,409	3,386
Private-non-voucher	554	459	445	445	442	432	435	427	423	419	422	426
<i>B. Participating schools</i>	–	–	–	–	6,503	6,864	6,912	7,011	7,150	7,133	7,111	7,125
Public	–	–	–	–	4,957	4,969	4,898	4,782	4,759	4,665	4,618	4,582
Private-voucher	–	–	–	–	1,546	1,895	2,014	2,229	2,391	2,468	2,493	2,543

Notes: Panel A reports the number of schools offering at least one grade from 1 to 8. Panel B reports the number of these schools participating in the targeted voucher program.

Tables 3–5 describe the analysis sample and the outcomes we study. We retain school-year observations where every outcome of interest is observed.

Table 3 reports means and standard deviations for annual school entry and exit in each municipality over the period 2005–2015. We define a local education market as a municipality. Entry and exit measure the number of schools that begin and cease operations, respectively, in each market. Panels A, B, and C report the statistics for public, private-voucher, and private-non-voucher schools.

Public school entry falls from an average of 0.16 schools per municipality in 2005 to 0.03 in 2015. Public school exits exceed entries in every year, averaging 0.25 per municipality in 2005 and

¹¹See Neilson (2013, 2020) and Sánchez (2026), among others.

0.17 in 2015. This pattern is consistent with the decline in the number of public schools shown in Table 2.

Entry is more frequent among private-voucher schools, particularly early in the period. The average number of entrants per municipality falls from 0.38 in 2005 to 0.10 in 2015. Entries exceed exits throughout 2005–2012. This pattern reverses in 2013, after which exits exceed entries for the remainder of the period.

Entry and exit are less frequent among private-non-voucher schools. Early in the period, exits generally exceed entries; in 2005, for example, there are 0.03 entries and 0.08 exits per municipality. From 2012 to 2015, both entry and exit average 0.02 schools per municipality per year.

Table 3: Summary Statistics - Entry and Exit

		2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015
<i>A. Public</i>												
	Entry	0.16 (0.50)	0.04 (0.21)	0.04 (0.23)	0.04 (0.25)	0.06 (0.26)	0.04 (0.20)	0.11 (0.34)	0.10 (0.34)	0.04 (0.22)	0.08 (0.29)	0.03 (0.25)
	Exit	0.25 (0.62)	0.41 (0.88)	0.26 (0.66)	0.23 (0.54)	0.18 (0.55)	0.26 (0.75)	0.46 (1.18)	0.27 (0.70)	0.29 (0.71)	0.31 (0.79)	0.17 (0.48)
<i>B. Private-voucher</i>												
	Entry	0.38 (0.86)	0.38 (0.87)	0.29 (0.67)	0.34 (0.80)	0.35 (0.77)	0.19 (0.53)	0.19 (0.52)	0.31 (0.69)	0.18 (0.61)	0.16 (0.45)	0.10 (0.38)
	Exit	0.21 (0.68)	0.15 (0.47)	0.19 (0.57)	0.18 (0.60)	0.17 (0.47)	0.09 (0.31)	0.12 (0.43)	0.21 (0.52)	0.22 (0.56)	0.17 (0.50)	0.16 (0.50)
<i>C. Private-non-voucher</i>												
	Entry	0.03 (0.19)	0.03 (0.23)	0.05 (0.34)	0.04 (0.23)	0.02 (0.19)	0.01 (0.11)	0.01 (0.08)	0.02 (0.13)	0.02 (0.15)	0.02 (0.17)	0.02 (0.14)
	Exit	0.08 (0.37)	0.04 (0.23)	0.03 (0.21)	0.04 (0.23)	0.02 (0.15)	0.01 (0.11)	0.03 (0.22)	0.02 (0.18)	0.02 (0.19)	0.02 (0.15)	0.02 (0.15)
	Observations	345	345	345	345	345	345	343	341	341	341	341

Notes: Entries and exits are measured as annual counts of schools per municipality. The table reports municipality-level means, with standard deviations in parentheses.

Table 4 reports means and standard deviations for six educational inputs over the period 2004–2015. These inputs capture schools’ efforts to improve educational quality. We examine the share of multigrade classes and class size, as well as the pupil-teacher ratio and weekly teaching hours per teacher. We also examine the shares of teachers holding indefinite contracts and of teachers with a specialization. As in Table 3, panels A, B, and C correspond to public, private-voucher, and private-non-voucher schools.

Educational inputs differ substantially across school types. This heterogeneity may reflect differences in objective functions, production functions, and school characteristics.¹² Despite these

¹²For example, public schools are more likely than private schools to be located in rural areas (Sánchez, 2026). Rural

differences, almost all inputs evolve in the direction of improving educational quality, especially after the introduction of the reform. For instance, the share of multigrade classes falls by roughly half in each sector over the study period. The only exception is the share of teachers holding indefinite contracts, which declines overall in all three sectors.

schools, in turn, are more likely to offer multigrade classes.

Table 4: Summary Statistics - Educational Inputs

	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015
<i>A. Public</i>												
multigrade classes (share)	0.46 (0.47)	0.45 (0.47)	0.45 (0.47)	0.45 (0.47)	0.44 (0.47)	0.44 (0.47)	0.40 (0.45)	0.31 (0.44)	0.34 (0.45)	0.23 (0.39)	0.24 (0.39)	0.26 (0.41)
class size	21.42 (11.10)	20.90 (11.08)	20.56 (11.07)	20.08 (11.03)	19.62 (10.99)	19.20 (10.97)	18.48 (11.07)	17.85 (11.42)	17.81 (11.14)	16.79 (11.84)	16.92 (11.69)	17.02 (11.49)
pupil-teacher ratio	16.46 (11.09)	15.68 (9.96)	15.52 (9.63)	14.80 (7.75)	14.59 (8.17)	14.32 (19.44)	13.74 (12.41)	12.90 (13.43)	11.45 (8.66)	10.70 (7.20)	10.00 (6.89)	9.56 (6.47)
indefinite contracts (share)	0.81 (0.24)	0.82 (0.23)	0.81 (0.23)	0.81 (0.23)	0.76 (0.25)	0.73 (0.26)	0.72 (0.27)	0.61 (0.29)	0.57 (0.29)	0.47 (0.29)	0.43 (0.29)	0.45 (0.28)
specialized teachers (share)	0.04 (0.10)	0.04 (0.10)	0.04 (0.11)	0.04 (0.10)	0.05 (0.11)	0.06 (0.12)	0.06 (0.12)	0.19 (0.20)	0.19 (0.21)	0.37 (0.28)	0.42 (0.28)	0.46 (0.28)
teaching hours per teacher	24.48 (4.98)	24.43 (5.09)	24.51 (5.17)	24.40 (5.16)	24.55 (5.28)	24.70 (5.39)	24.70 (5.54)	22.66 (5.87)	22.63 (5.78)	21.81 (5.75)	21.48 (5.74)	20.84 (5.72)
Observations	5,411	5,375	5,253	5,176	5,100	5,052	4,971	4,758	4,777	4,713	4,638	4,601
<i>B. Private-voucher</i>												
multigrade classes (share)	0.23 (0.40)	0.23 (0.39)	0.22 (0.39)	0.22 (0.39)	0.21 (0.39)	0.20 (0.38)	0.19 (0.37)	0.15 (0.33)	0.14 (0.33)	0.09 (0.27)	0.10 (0.28)	0.11 (0.29)
class size	28.35 (11.07)	27.81 (11.07)	27.74 (11.05)	27.70 (10.87)	27.64 (11.06)	27.59 (10.81)	27.17 (11.01)	26.48 (11.46)	26.51 (11.32)	25.77 (12.34)	25.97 (12.14)	26.24 (11.96)
pupil-teacher ratio	21.37 (12.35)	20.06 (10.20)	20.08 (9.45)	19.83 (9.83)	19.62 (9.29)	19.71 (13.38)	18.74 (7.94)	17.24 (7.86)	17.20 (12.37)	15.99 (7.19)	15.51 (8.06)	15.18 (6.96)
indefinite contracts (share)	0.77 (0.27)	0.76 (0.27)	0.75 (0.27)	0.74 (0.27)	0.73 (0.27)	0.73 (0.27)	0.73 (0.26)	0.68 (0.26)	0.64 (0.27)	0.59 (0.27)	0.56 (0.26)	0.56 (0.26)
specialized teachers (share)	0.15 (0.18)	0.16 (0.18)	0.16 (0.18)	0.16 (0.18)	0.17 (0.18)	0.18 (0.19)	0.19 (0.19)	0.29 (0.22)	0.29 (0.22)	0.46 (0.25)	0.50 (0.25)	0.53 (0.25)
teaching hours per teacher	21.39 (5.66)	21.36 (5.62)	21.34 (5.59)	21.30 (5.61)	21.35 (5.52)	21.30 (5.46)	21.26 (5.46)	20.14 (5.51)	20.22 (5.47)	19.30 (5.39)	19.03 (5.23)	18.40 (5.18)
Observations	2,853	3,013	3,106	3,143	3,201	3,247	3,283	3,341	3,369	3,397	3,395	3,374
<i>C. Private-non-voucher</i>												
multigrade classes (share)	0.02 (0.11)	0.02 (0.10)	0.02 (0.11)	0.02 (0.10)	0.02 (0.11)	0.01 (0.09)	0.01 (0.08)	0.01 (0.10)	0.01 (0.08)	0.01 (0.06)	0.00 (0.06)	0.01 (0.06)
class size	20.55 (9.00)	20.49 (8.66)	20.75 (8.80)	20.99 (8.49)	21.02 (8.37)	22.49 (10.94)	21.23 (8.24)	21.83 (7.76)	21.00 (8.18)	21.87 (11.72)	21.59 (8.08)	21.74 (8.21)
pupil-teacher ratio	13.21 (11.28)	11.75 (6.49)	11.79 (6.29)	12.09 (6.34)	12.12 (6.17)	12.53 (7.19)	12.46 (6.65)	15.52 (32.73)	11.95 (8.26)	11.59 (5.36)	12.00 (20.40)	25.02 (58.22)
indefinite contracts (share)	0.82 (0.22)	0.84 (0.2)	0.84 (0.2)	0.82 (0.21)	0.82 (0.19)	0.81 (0.21)	0.81 (0.21)	0.84 (0.19)	0.81 (0.22)	0.76 (0.23)	0.73 (0.24)	0.71 (0.24)
specialized teachers (share)	0.36 (0.21)	0.38 (0.20)	0.39 (0.20)	0.38 (0.20)	0.39 (0.21)	0.40 (0.20)	0.41 (0.21)	0.50 (0.19)	0.48 (0.20)	0.58 (0.18)	0.63 (0.18)	0.66 (0.16)
teaching hours per teacher	17.98 (5.26)	17.69 (5.48)	17.68 (5.21)	17.72 (4.97)	17.7 (4.94)	17.67 (5.13)	17.68 (5.08)	17.18 (5.48)	17.06 (5.16)	15.98 (4.67)	15.33 (4.38)	14.63 (4.86)
Observations	541	458	443	436	434	421	422	323	415	416	420	413

Notes: The table reports school-level means, with standard deviations in parentheses. Teaching hours per teacher are measured per week.

Table 5 reports mean top-up fees and their standard deviations for private-voucher schools over the period 2004–2015. Fees are expressed in real US dollars, using the May 16, 2016 exchange rate. Before the reform, each school could charge a single top-up fee to all students, regardless of their socioeconomic status. This fee could be zero or positive.

The targeted voucher program introduced a form of third-degree price discrimination at participating schools. These schools were required to charge eligible low-income students zero fees, while remaining able to charge positive fees to higher-income students. Non-participating schools could continue to charge the same fee to all students. Thus, from 2008 onward, we distinguish two vectors of fees in each market: one faced by eligible low-income students and another faced by non-eligible higher-income students. The averages give each school equal weight and include the mandated zero fees faced by eligible students at participating schools.

The average fee before the reform is the same for both groups, rising from \$16.14 in 2004 to \$17.24 in 2007. After the reform, the average fee faced by eligible low-income students falls to approximately \$14–15. The average fee faced by higher-income students rises to \$20.04 by 2015. These figures describe the evolution of average fees across schools. Our event-study analysis in Section 4 uses differences in treatment intensity across markets to estimate the reform’s effects.

Table 5: Summary Statistics - Private-voucher School Top-up Fees

	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015
Low income	16.14 (25.29)	17.36 (26.81)	18.34 (31.25)	17.24 (26.92)	14.46 (27.31)	14.97 (34.07)	15.04 (29.86)	14.40 (30.05)	14.96 (35.24)	15.41 (37.28)	14.95 (33.32)	14.71 (34.31)
Higher income	16.14 (25.29)	17.36 (26.81)	18.34 (31.25)	17.24 (26.92)	17.82 (27.47)	19.33 (34.36)	19.29 (30.39)	18.98 (30.42)	20.49 (36.67)	20.87 (37.36)	20.13 (33.51)	20.04 (34.67)
Observations	2,853	3,013	3,106	3,143	3,201	3,247	3,283	3,341	3,369	3,397	3,395	3,374

Notes: The table reports school-level mean fees for each eligibility group, with standard deviations in parentheses. Each school receives equal weight, and fees for eligible low-income students are zero at participating schools. All amounts are expressed in real US dollars, converted from Chilean pesos at the May 16, 2016 exchange rate of Ch\$686.52 per US dollar.

4 Empirical Analysis

4.1 Identification Strategy

We define local education markets using municipal boundaries, following Hsieh and Urquiola (2006) and Sánchez (2019).¹³ Enrollment patterns support this definition: about 90% of students attend school in their municipality of residence.

¹³See Topel (1986) and Card (2001) for other studies that use political and administrative boundaries to define local markets.

The design of the targeted voucher program creates differences in exposure across markets. A municipality with no eligible low-income students cannot receive additional funds through the targeted voucher. At the other extreme, a municipality in which all students are eligible has the greatest potential exposure. We therefore measure treatment intensity by the municipality's share of eligible low-income students. This measure captures ex-ante exposure because the additional funds a market actually receives also depend on schools' decisions to participate. As we show below, the eligibility share is a strong predictor of private-voucher schools' participation.

We construct treatment intensity using students' municipality of residence in 2007, the year before the program was introduced. This choice avoids endogeneity arising from residential migration in response to the reform.¹⁴ The 2007 municipality of residence is highly correlated with students' subsequent residence.¹⁵ This pre-reform measure is free of endogeneity because residential choices preceded the program's announcement and implementation.

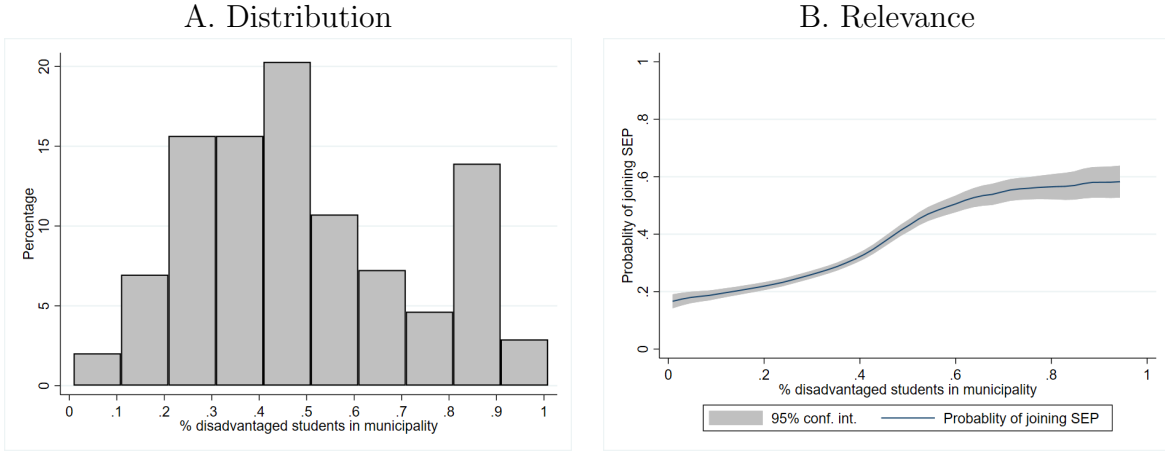
Figure 1 presents the distribution and relevance of treatment intensity. Panel A shows that the distribution spans the full $[0, 1]$ range. About half of municipalities have eligibility shares between 20% and 50%, while few have shares below 10% or above 90%. This distribution provides substantial variation in exposure to the reform.

Panel B plots a nonparametric estimate of the probability that a private-voucher school joins the program in 2008 against its municipality's eligibility share. The estimated probability rises monotonically from approximately 0.2 to 0.6 over the $[0, 1]$ range. Thus, the pre-reform eligibility share both varies substantially across markets and strongly predicts schools' participation in the program.

¹⁴Current residence may reflect endogenous migration, as discussed by Rosenzweig and Wolpin (1988).

¹⁵Among fourth graders observed in 2008–2011, 88% live in the same municipality as in 2007.

Figure 1: Distribution and Relevance of Intensity of Treatment



Notes: Panel A shows the distribution of municipalities by their share of eligible low-income students, using students' municipality of residence in 2007. Panel B shows a nonparametric estimate of the probability that a private-voucher school participates in the program in 2008 as a function of this share.

4.2 Empirical Strategy

We estimate the reform's effects using an event-study design. The design exploits the unanticipated introduction of the program in 2008 and exogenous differences in pre-reform treatment intensity across municipalities. We examine the outcomes described in Tables 3–5.

For school entry and exit, we estimate

$$y_{st} = \gamma_s + \lambda_t + \sum_{\substack{\tau=-3 \\ \tau \neq -1}}^7 \delta_\tau (D_{\tau t} \times \text{intensity}_s) + \varepsilon_{st}. \quad (1)$$

The outcome y_{st} is the number of schools entering or exiting municipality s in year t . The terms γ_s and λ_t are municipality and year fixed effects, respectively. The variable $D_{\tau t}$ indicates that year t is τ years from the reform, with $\tau = 0$ corresponding to 2008. Treatment intensity, intensity_s , is the share of eligible low-income students assigned to municipality s using their pre-reform residence. We omit the interaction for $\tau = -1$, making 2007 the reference year. The error term is ε_{st} . We cluster standard errors at the municipality level (Bertrand et al., 2004; Cameron and Miller, 2015).

For educational inputs and top-up fees, we estimate the corresponding school-level specification:

$$y_{ist} = \gamma_s + \lambda_t + \sum_{\substack{\tau=-4 \\ \tau \neq -1}}^7 \delta_\tau (D_{\tau t} \times \text{intensity}_s) + \varepsilon_{ist}. \quad (2)$$

Here, y_{ist} is the outcome for school i in municipality s and year t . The fixed effects remain at the municipality and year levels. Each school-year observation receives equal weight, and standard errors are again clustered at the municipality level.

The coefficients of interest are δ_τ . For post-reform periods, our design identifies average treatment effects on the treated, with treatment defined as full relative to null exposure to the reform. Full treatment corresponds to a municipality in which all students are eligible, while null treatment corresponds to one in which none are eligible. The coefficients measure the effect of this contrast relative to 2007.

Because treatment is defined at the market level, the estimates incorporate equilibrium responses on both sides of the market. These include student sorting across schools on the demand side and schools' participation decisions on the supply side. The estimates also include the responses of both participating and non-participating schools.¹⁶ This treatment differs from that studied by Correa et al. (2014) and Feigenberg et al. (2019), who examine the effect of a school choosing to participate in the program. Their treatment does not necessarily capture equilibrium effects across the market.

4.3 Results

We present the event-study results for entry and exit, educational inputs, and top-up fees in figures. The discussion focuses on public and private-voucher schools. Estimates for private-non-voucher schools are imprecise and generally not statistically distinguishable from zero. Tables B.1–B.8 in Appendix B report the point estimates and standard errors for all three sectors. Throughout the discussion, we use 2007 sample means to benchmark the magnitudes of the estimated full-versus-null treatment effects.

4.3.1 Entry and Exit

Figure 2 reports the effects on school entry in panel A and school exit in panel B. A school enters in year t if it is observed in that year but not in $t - 1$. It exits in year t if it is observed in $t - 1$ but not in t . Blue circles represent public schools, and red diamonds represent private-voucher

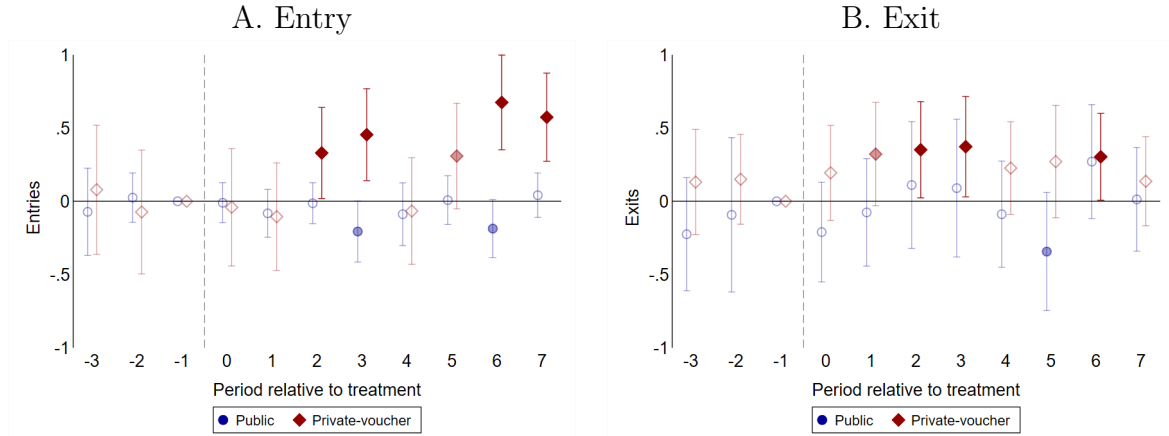
¹⁶In an oligopolistic market, schools' strategies impose externalities on their competitors. The reform can therefore affect the decisions of schools that participate and those that do not. See the model in Sánchez (2026).

schools. The figure includes 95% confidence intervals based on standard errors clustered at the municipality level. We find no statistically significant pre-trends, confirming the internal validity of our design.

Panel A shows little response in public school entry. The estimated effects are significant only at the 10% level in periods 3 and 6, when entry falls by approximately 0.2 schools per municipality. Private-voucher schools respond more strongly, although the response is not immediate. The effect first becomes statistically significant in period 2, with approximately 0.3 additional entrants per municipality, and is larger in period 3. The largest effects occur in periods 6 and 7, at approximately 0.6 additional entrants per municipality. These late-period effects represent more than 100% of the average annual entry count in 2007 (Table 3).

Panel B shows a similar contrast for school exit. Public schools respond little and, if anything, exit less often. None of the estimated effects is statistically significant at the 5% level. Private-voucher schools increase their exit frequency by approximately 0.3 schools per municipality per year. The effects are statistically significant in periods 1, 2, 3, and 6. Their magnitude is substantial relative to the average of 0.19 private-voucher school exits per municipality in 2007.

Figure 2: Effects of the Targeted Voucher Program on Public and Private-voucher Schools - Entry and Exit



Notes: A school enters in year t if it is observed in t but not in $t - 1$, and exits in t if it is observed in $t - 1$ but not in t . Entry and exit are constructed from school data for 2004–2015; the regressions use municipality-level observations for 2005–2015. Coefficients compare full with null treatment, relative to 2007. Event time zero corresponds to 2008. Point estimates and 95% confidence intervals are shown, with standard errors clustered at the municipality level. Public schools are represented by blue circles and private-voucher schools by red diamonds. Hollow markers denote statistical insignificance at the 10% level. Light-colored markers denote significance at the 10% level but not at the 5% level, and dark-colored markers denote significance at the 5% level.

In sum, public schools do not modify their entry and exit behavior in response to the reform, while private-voucher schools respond relatively quickly and in sizable numbers. This contrast is consistent with the arguments in Pritchett (2013). He attributes public schools’ slow adjustment to the limited flexibility of the top-down bureaucracies that administer them. Private schools, by comparison, face stronger market incentives and make decisions at the institution level, allowing them to respond more rapidly to changes in policy and regulation.

4.3.2 Educational Inputs

Figure 3 presents the effects on six educational inputs. Panels A and B examine infrastructure-related inputs: the share of multigrade classes and class size. Panels C–F examine the pupil-teacher ratio, the share of teachers holding indefinite contracts, the share of specialized teachers, and weekly teaching hours per teacher. Pre-trends are parallel for the vast majority of outcomes, again confirming the internal validity of our design.

Panel A shows that both public and private-voucher schools reduce their share of multigrade classes. The response takes time to emerge, becoming statistically significant in period 3 in both sectors. The effects are economically large, especially toward the end of the study period. By period 7, the estimated reduction for public schools exceeds 80% of their 2007 mean, while the reduction for private-voucher schools is approximately 220% of their 2007 mean (Table 4). These percentages benchmark the full-versus-null treatment coefficients against the observed pre-reform means.

Panel B shows that the reform reduces class size, particularly in private-voucher schools. Public schools’ response moves in the direction of improving quality but is comparatively small. Statistically significant effects appear only in periods 5 and 6, with reductions of approximately 1–2 students per class, or 5–10% of the 2007 mean. Private-voucher schools reduce class size beginning in period 1, and the effects remain statistically significant thereafter. The largest reductions occur in periods 5 and 6, at approximately nine students per class.

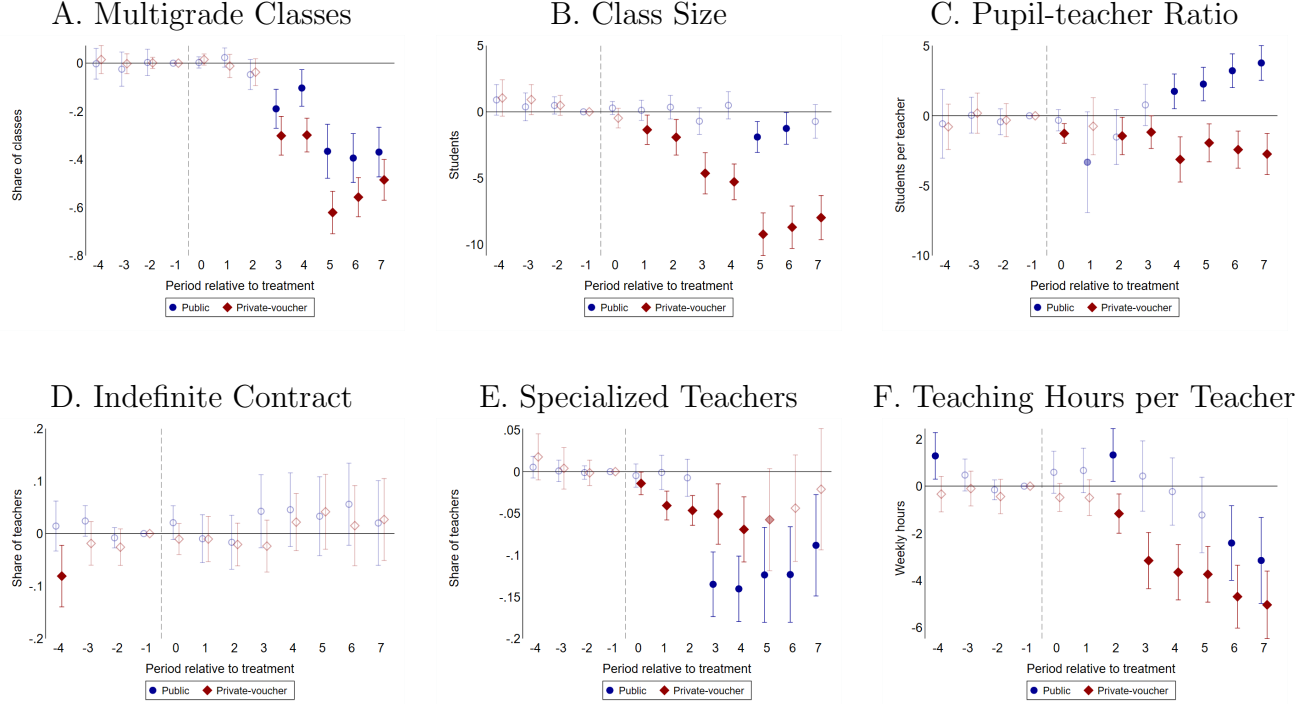
The responses of teaching-staff inputs generally also move in the direction of improving quality, especially in private-voucher schools. Panel C shows an important difference between sectors: public schools increase their pupil-teacher ratio, while private-voucher schools reduce it. The increases in public schools become statistically significant late in the period and amount to 10–25% of the 2007 mean. Private-voucher schools respond immediately, reducing the ratio by approximately one student per teacher in the year of implementation. By the end of the period, the estimated reduction is approximately three students per teacher.

Panel D shows no statistically significant effect on the share of teachers holding long-term indefinite contracts in either sector. This finding suggests that the reform did not affect teachers’ job stability.

Panel E reports the effects on the share of specialized teachers. These teachers have additional training in one or more subjects and are therefore expected to have greater teaching skills than otherwise comparable teachers without that training. Both sectors reduce their share of specialized teachers at some point after the reform, but the reduction is larger in public schools. The public-school effects become statistically significant in period 3 and amount to approximately 225–350% of the 2007 mean. Private-voucher schools respond earlier, but their reductions are smaller and reverse by the end of the period. The largest private-voucher school effect occurs in period 4 and is approximately 5% of the 2007 mean. The private-voucher school effects are no longer statistically significant by the end of the study period.

Finally, panel F examines weekly teaching hours per teacher. Longer hours in front of students may increase fatigue and impair teaching performance. Both sectors ultimately reduce teaching hours, but public schools take longer to respond. After a small increase in period 2, public schools reduce weekly teaching hours by approximately 2–3 hours in periods 6 and 7. Private-voucher schools show larger reductions, with negative and statistically significant effects throughout periods 2–7. These reductions range from approximately 1.2 to five hours per week.

Figure 3: Effects of the Targeted Voucher Program on Public and Private-voucher Schools - Educational Inputs



Notes: The event-study regressions use school-level data for 2004–2015. Each panel reports effects on a different educational input. Coefficients compare full with null treatment, relative to 2007. Event time zero corresponds to 2008. Point estimates and 95% confidence intervals are shown, with standard errors clustered at the municipality level. Public schools are represented by blue circles and private-voucher schools by red diamonds. Hollow markers denote statistical insignificance at the 10% level. Light-colored markers denote significance at the 10% level but not at the 5% level, and dark-colored markers denote significance at the 5% level.

Overall, both public and private-voucher schools respond to the reform by adjusting educational inputs. Schools generally improve their infrastructure-related inputs and inputs associated with teaching staff. Consistent with Pritchett (2013), public schools take longer to adjust than private-voucher schools. Nevertheless, public schools eventually respond, and their adjustments mostly move in the direction of increasing educational quality (Hoxby, 2003b).

4.3.3 Top-up Fees

The reform introduced a form of third-degree price discrimination at participating private-voucher schools. Before the reform, each school could charge the same top-up fee to all students. After the reform, participating schools had to charge eligible low-income students zero fees, while remaining

able to charge positive fees to higher-income students. Non-participating schools retained the same pricing rule for all students.

We therefore distinguish two vectors of fees in each market. Eligible low-income students face zero fees at participating schools and the posted fee at non-participating schools. Higher-income students face the posted fee at every school, regardless of its participation status. Our school-level outcomes record the fee available to each group, with each school-year observation receiving equal weight.

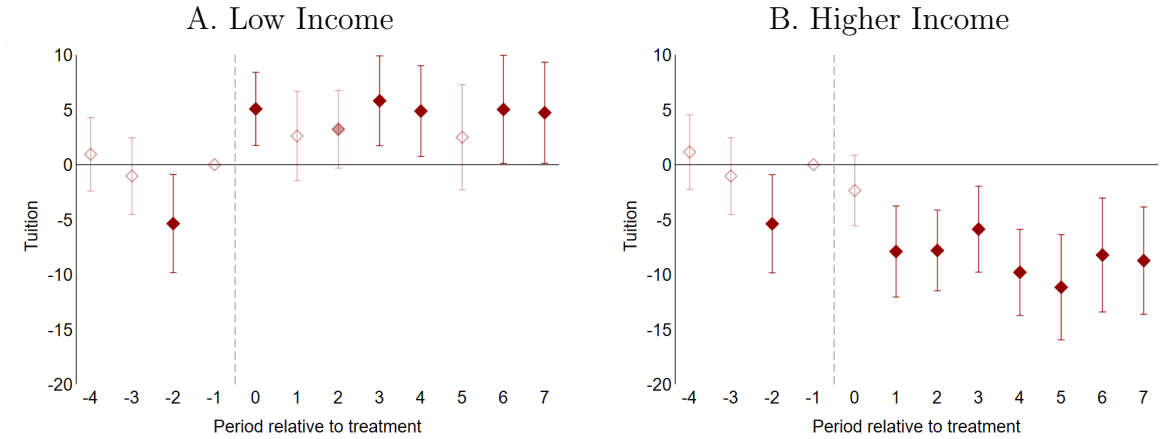
Figure 4 presents the event-study estimates for eligible low-income students in panel A and higher-income students in panel B.¹⁷ Pre-reform coefficients are generally not statistically distinguishable from zero, with the exception of period -2 . As in the preceding results, the coefficients compare full with null treatment relative to 2007.

Panel A shows that full relative to null treatment increases the average fee faced by eligible low-income students. The response is immediate and statistically significant in the year of implementation. The point estimates remain positive and fairly stable over the study period, at approximately \$3–6. This finding is particularly surprising given the requirement that participating schools charge eligible students zero fees.

Panel B shows that full relative to null treatment reduces the average fee faced by higher-income students. The response is quick and economically large. Estimated reductions of approximately \$8–11 correspond to 45–65% of the average fee charged in 2007.

¹⁷Table B.8 in Appendix B reports the corresponding point estimates and standard errors.

Figure 4: Effects of the Targeted Voucher Program on Private-voucher Schools - Top-up Fees



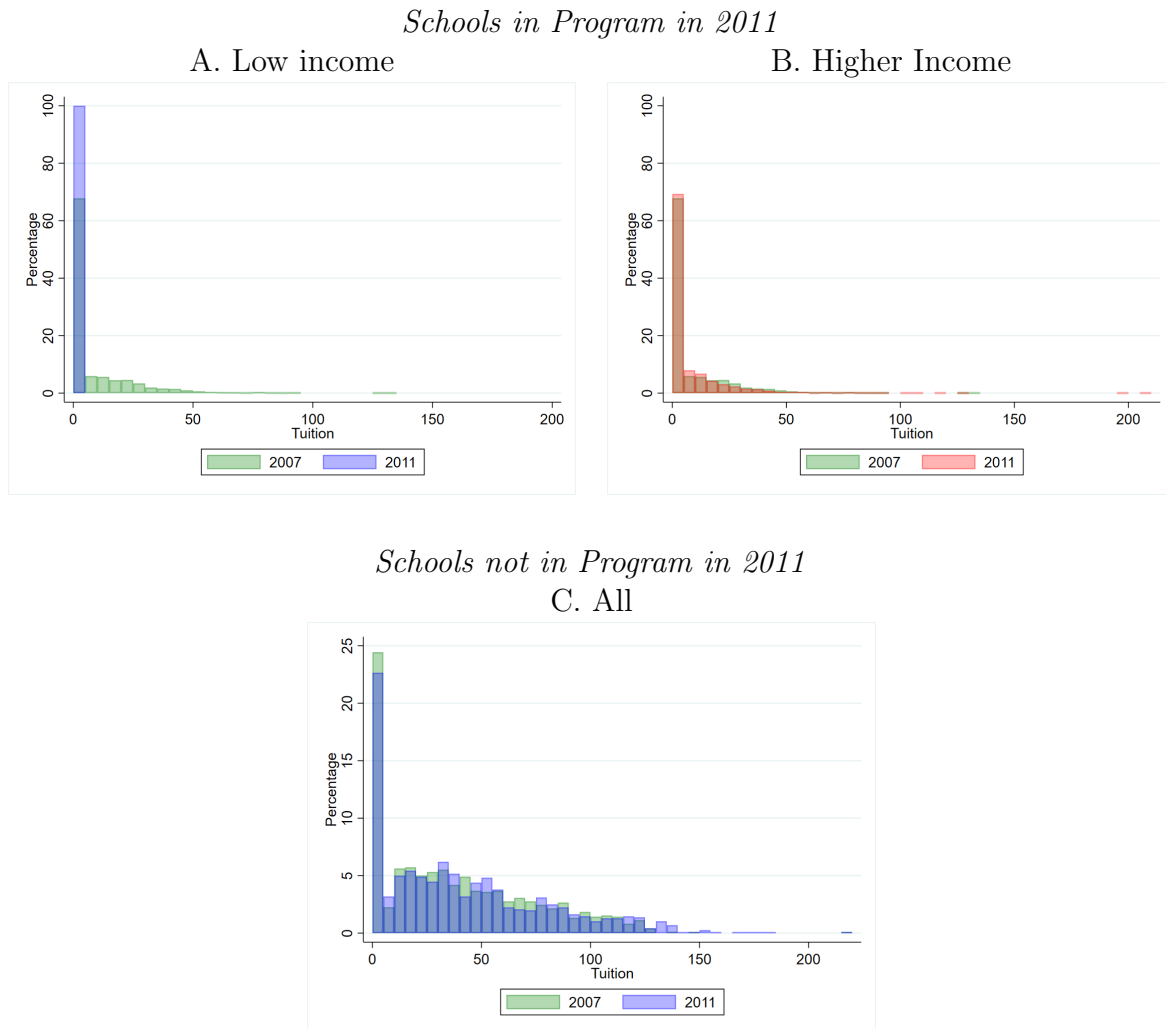
Notes: Fees are expressed in real US dollars, converted from Chilean pesos at the May 16, 2016 exchange rate of Ch\$686.52 per US dollar. The event-study regressions use school-level data for 2004–2015, with each school-year observation receiving equal weight. Coefficients compare full with null treatment, relative to 2007. Event time zero corresponds to 2008. Point estimates and 95% confidence intervals are shown, with standard errors clustered at the municipality level. Hollow diamonds denote statistical insignificance at the 10% level. Light-colored diamonds denote significance at the 10% level but not at the 5% level, and dark-colored diamonds denote significance at the 5% level.

Figures 5 and 6 describe fee distributions and average fees separately by schools' participation status in 2011. We use this reference year as a descriptive way to distinguish schools more inclined to participate in the program from those less inclined to do so. The classification is held fixed across the years shown.

Figure 5 compares the distributions in 2007, the year before the reform, and 2011, the program's fourth year. Panels A and B concern schools participating in 2011. Panel A shows that these schools generally charged low fees to eligible low-income students before the reform, with approximately 70% charging zero fees in 2007. By 2011, all schools in this group charged eligible students zero fees, as required by the program.

Panel B shows the fees faced by higher-income students at the same group of schools. The 2007 and 2011 distributions resemble each other, but the 2011 distribution is shifted to the left, indicating lower fees in that year. Panel C presents the distributions for schools not participating in 2011, which charge the same fee to both student groups. Only approximately 23% of these schools charge zero fees in 2007, compared with 25% in 2011. Although the distributions are similar, the 2011 distribution is shifted to the right, indicating higher fees.

Figure 5: Private-voucher Schools' Monthly Top-up Fees Distribution - Before and After the Introduction of the Targeted Voucher Program



Notes: The figure compares fee distributions in 2007 and 2011. Schools are grouped by their participation status in 2011. Panels A and B show fees for eligible low-income and higher-income students, respectively, at schools participating in 2011. Panel C shows fees for all students at schools not participating in 2011. Fees are expressed in real US dollars, converted from Chilean pesos at the May 16, 2016 exchange rate of Ch\$686.52 per US dollar.

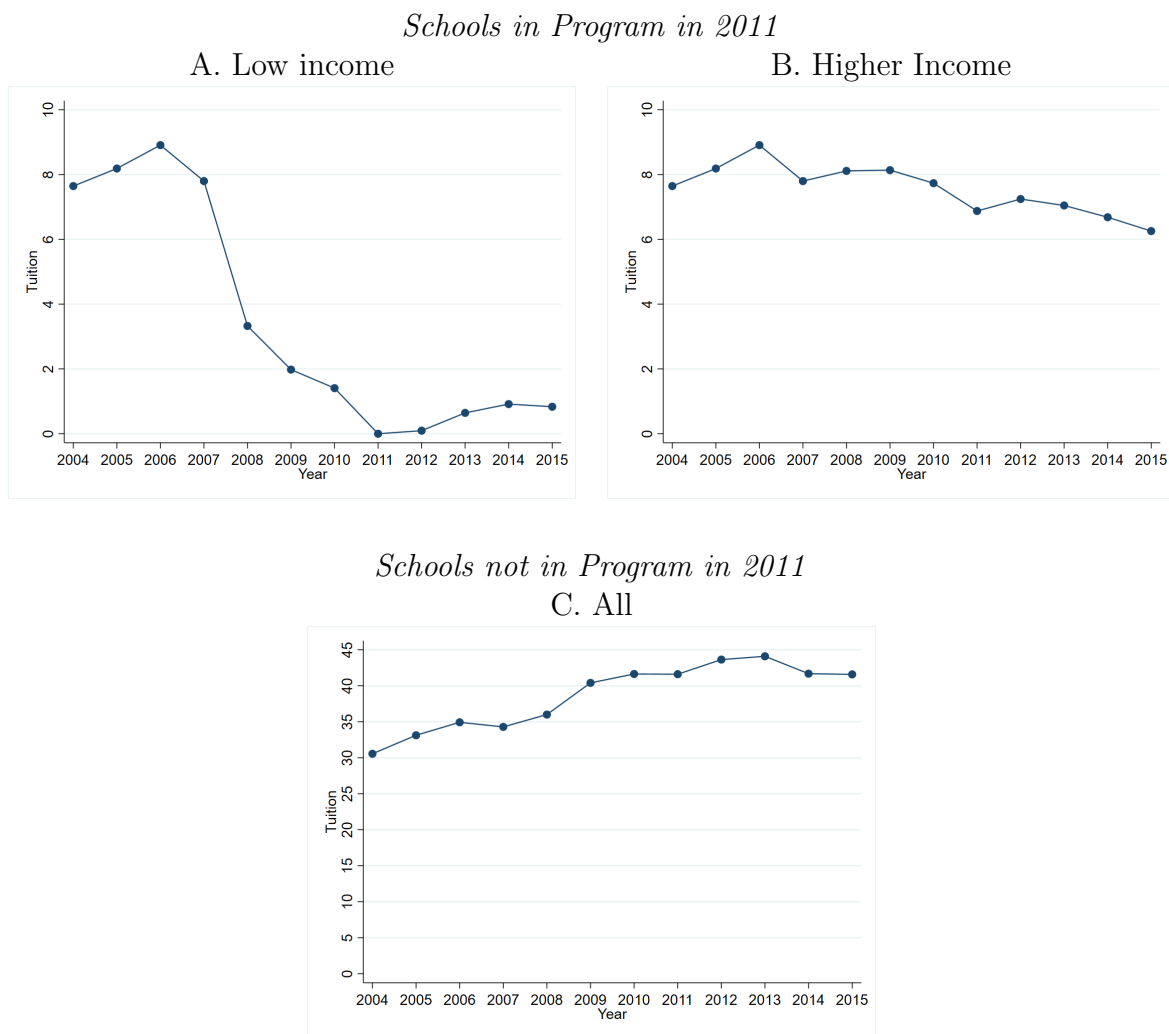
Figure 6 presents the evolution of average fees for the same school groups. Among schools participating in 2011, average fees faced by eligible low-income students fall sharply after the reform, from approximately \$8 in 2007 to nearly zero in 2015 (panel A). Eligible students face zero fees at every school in this group in 2011, when all of these schools participate in the program.¹⁸

¹⁸The average is not exactly zero in every post-reform year because the groups are defined by participation in 2011.

Average fees faced by higher-income students at these schools decline steadily from approximately \$8 in 2007 to \$6 in 2015 (panel B).

Among schools not participating in 2011, average fees rise steadily both before and after the reform (panel C). They increase from approximately \$30 in 2004 to \$35 in 2007 and \$42 in 2015.

Figure 6: Private-voucher Schools' Average Monthly Top-up Fees over Time



Notes: The figure shows school-level mean fees over time, grouping schools by their participation status in 2011. Panels A and B show fees for eligible low-income and higher-income students, respectively, at schools participating in 2011. Panel C shows fees for all students at schools not participating in 2011. Schools' participation status in other years may differ from their status in 2011. Fees are expressed in real US dollars, converted from Chilean pesos at the May 16, 2016 exchange rate of Ch\$686.52 per US dollar.

Some schools in this group joined after 2008, while others left the program after 2011.

In sum, the event-study estimates show that full relative to null treatment raises the average fee faced by eligible low-income students and lowers the average fee faced by higher-income students. The increase for eligible students occurs despite the requirement that participating schools charge them zero fees.

5 Conclusions

We examine school responses to the implementation of a recent targeted voucher program in Chile. Our results show that public and (especially) subsidized private schools respond to the reform along various margins. Private schools enter and exit more often the market than in the absence of the reform. Schools make successful efforts in improving infrastructure related inputs. When it comes to educational inputs that are related to schools' teaching staff, in general schools' responses are in the direction of improving quality. Lastly, private schools' top-up fee responses are such that low income students see the average fee they face in the market increase, while higher income students see the average fee they face in the market decrease.

Possibly the most interesting result we find is the one on private-voucher schools' fees. Surprisingly, the reform induces schools to increase the top-up fee that is charged to low income students, even with the reform's mandate of participating schools charging zero top-up fees to eligible low income students. We examine the mathematical conditions for our results on fees to occur, and conclude that the counterfactual fees schools charge in the case they do not join the program under the reform need to be sufficiently higher than their fees in the absence of the reform, and that the counterfactual fees schools charge to higher income students in the case they participate in the program under the reform need to be sufficiently lower than their fees in the absence of the reform. We show descriptive statistics supporting these conditions.

Our paper greatly adds to the educational vouchers literature. In particular, it complements existing evidence on the effects of Chile's recent targeted voucher reform, by investigating a piece of the puzzle (supply side) that is rarely the focus of study, but that is central to explaining and producing the observed effects on students' achievement. Furthermore, policymakers wanting to evaluate existing voucher programs or considering implementing voucher policies are in great need of understanding the supply side responses that are triggered by this type of program. Effective designs of voucher programs take into account all of the consequences these programs have on school markets; in particular, supply side responses (Epple et al., 2017).

References

- ABDULKADIROGLU, A., P. A. PATHAK, AND C. R. WALTERS (2018): “Free to Choose: Can School Choice Reduce Student Achievement?” *American Economic Journal: Applied Economics*, 10, 175–206.
- AGUIRRE, J. (2020): “How can progressive vouchers help the poor benefit from school choice? Evidence from the Chilean voucher system,” *Journal of Human Resources*.
- ALLENDE, C. (2019): “Competition Under Social Interactions and the Design of Education Policies,” Working paper.
- ALLENDE, C., F. GALLEGGO, AND C. NEILSON (2019): “Approximating the Equilibrium Effects of Informed School Choice,” Working paper.
- BERTRAND, M., E. DUFLO, AND S. MULLAINATHAN (2004): “How much should we trust differences-in-differences estimates?” .
- CAMERON, C. A. AND D. L. MILLER (2015): “A Practitioner’s Guide to Cluster-Robust Inference,” *Journal of Human Resources*.
- CARD, D. (1992): “Using Regional Variation in Wages to Measure the Effects of the Federal Minimum Wage,” *Industrial and Labor Relations Review*, 46, 22–37.
- (2001): “Immigrant Inflows, Native Outflows, and the Local Labor Market Impacts of Higher Immigration,” *Journal of Labor Economics*, 19, 22–64.
- CHAKRABARTI, R. (2008): “Can increasing private school participation and monetary loss in a voucher program affect public school performance? Evidence from Milwaukee,” *Journal of Public Economics*, 92, 1371–1393.
- (2013a): “Accountability with Voucher Threats, Responses, and the Test-Taking Population: Regression Discontinuity Evidence from Florida,” *Education Finance and Policy*, 8, 121–167.
- (2013b): “Vouchers, Public School Response, and the Role of Incentives: Evidence from Florida,” *Economic Inquiry*, 51, 500–526.
- CORREA, J. A., F. PARRO, AND L. REYES (2014): “The effects of vouchers on school results: Evidence from Chile’s targeted voucher program,” *Journal of Human Capital*.

- DINERSTEIN, M., C. NEILSON, AND S. OTERO (2020): “The Equilibrium Effects of Public Provision in Education Markets: Evidence from a Public School Expansion Policy,” Working paper.
- DINERSTEIN, M. AND T. D. SMITH (2021): “Quantifying the Supply Response of Private Schools to Public Policies,” *American Economic Review*, 111, 3376–3417.
- DUFLO, E. (2001): “Schooling and Labor Market Consequences of School Construction in Indonesia: Evidence from an Unusual Policy Experiment,” *American Economic Review*, 91, 795–813.
- EPPLE, D., R. E. ROMANO, AND M. URQUIOLA (2017): “School Vouchers: A Survey of the Economics Literature,” *Journal of Economic Literature*, 55, 441–92.
- FEIGENBERG, B., R. YAN, AND S. RIVKIN (2019): “Illusory Gains from Chile’s Targeted School Voucher Experiment,” *The Economic Journal*, 129, 2805–2832.
- FERREYRA, M. M. AND G. KOSENOK (2018): “Charter school entry and school choice: The case of Washington, D.C.” *Journal of Public Economics*, 159, 160–182.
- FIGLIO, D. AND C. M. D. HART (2014): “Competitive Effects of Means-Tested School Vouchers,” *American Economic Journal: Applied Economics*, 6, 133–56.
- FIGLIO, D. N. AND C. E. ROUSE (2006): “Do accountability and voucher threats improve low-performing schools?” *Journal of Public Economics*, 90, 239–255.
- GAZMURI, A. M. (2015): “School Segregation in the Presence of Student Sorting and Cream-Skimming: Evidence from a School Voucher Reform,” Working paper.
- HOXBY, C. (2003a): “School choice and school competition: Evidence from the United States,” *Swedish Economic Policy Review*, 10, 9–65.
- (2003b): “School Choice and School Productivity (or Could School Choice be a Tide that Lifts All Boats?),” in *The Economics of School Choice*, University of Chicago Press, chap. 8, 287–341.
- HSIEH, C.-T. AND M. URQUIOLA (2006): “The effects of generalized school choice on achievement and stratification: Evidence from Chile’s voucher program,” *Journal of Public Economics*, 90, 1477–1503.
- LUCAS, A. M. AND I. M. MBITI (2012): “Access, Sorting, and Achievement: The Short-Run Effects of Free Primary Education in Kenya,” *American Economic Journal: Applied Economics*, 4, 226–53.

- NAVARRO-PALAU, P. (2017): “Effects of differentiated school vouchers: Evidence from a policy change and date of birth cutoffs,” *Economics of Education Review*, 58, 86–107.
- NEILSON, C. (2013): “Targeted Vouchers, Competition Among Schools, and the Academic Achievement of Poor Students,” PhD Dissertation, Yale University.
- (2020): “Targeted Vouchers, Competition Among Schools, and the Academic Achievement of Poor Students,” Unpublished manuscript.
- PRITCHETT, L. (2013): *The rebirth of education: schooling ain’t learning*, Center for Global development.
- ROSENZWEIG, M. AND K. I. WOLPIN (1988): “Migration selectivity and the effects of public programs,” *Journal of Public Economics*, 37, 265–289.
- SÁNCHEZ, C. (2019): “Skipping your Exam? The Unexpected Response to a Targeted Voucher Policy,” Working paper.
- (2026): “Equilibrium Consequences of Vouchers Under Simultaneous Extensive and Intensive Margins Competition,” Working paper.
- SINGLETON, J. D. (2019): “Incentives and the Supply of Effective Charter Schools,” *American Economic Review*, 109, 2568–2612.
- TOPEL, R. (1986): “Local Labor Markets,” *Journal of Political Economy*, 94, S111–43.

A Data

We combine various administrative data sets for Chilean students and schools for the years 2004–2015. Specifically, we use:

- Registry of students, 2004–2015.

These data provide information on students’ gender, date of birth, age, municipality of residence, type and level of education, grade, class, grade repetition status, special education status, and various characteristics of the school of attendance, such as municipality, type of administration (public, private-voucher, private-non-voucher), single/double shift schedule, and urban status.

- Registry of students’ academic performance, 2004–2015.

These data provide information on students’ gender, date of birth, municipality of residence, type and level of education, grade, class, GPA, average class attendance, and various characteristics of the school of attendance, such as municipality, type of administration, and urban status.

- Registry of schools, 2004–2015.

These data provide information on schools’ municipality, type of administration, urban status, address, and type and level of education offered.

- Registry of schools’ summary of enrollment, 2004–2015.

These data provide information on schools’ municipality, type of administration, urban status, male enrollment by education type and level, female enrollment by education type and level, total enrollment by education type and level, total enrollment, number of single-grade classes by education type and level, total number of single-grade classes, number of multigrade classes by education type and level, and total number of multigrade classes.

- Registry of teachers, 2004–2015.

These data provide information on teachers’ gender, date of birth, education degree, subject specialization, institution attended, graduation year, and duration of the degree studied. They also provide information on the characteristics of all schools in which each teacher is hired (municipality, type of management, rural status), and on the teachers’ primary and secondary roles (e.g. teacher, principal, supervisor), type of contract, hours contracted, teaching hours, experience, tenure, and teaching subject and level of education.

- Registry of schools that participate in the targeted voucher program, 2008–2015.

These data provide information on the characteristics of schools that participate in the targeted voucher program. Information on schools’ municipality, type of administration, urban

status, targeted voucher classification, number of low income students that are eligible to benefit from the targeted voucher subsidy, and number of targeted voucher beneficiary students is also available.

- Registry of students that are eligible to participate in the targeted voucher program, 2008–2015.

These data provide information on the characteristics of low income students that are eligible to be beneficiaries of the targeted voucher program. They provide information on students' gender, date of birth, targeted voucher participation status, level of education, grade, single/double shift schedule, and on the type of administration, urban status, and targeted voucher category of the school attended by the student.

- Private-voucher schools' top-up fees, 2004–2015.

These databases list all private-voucher schools that charge strictly positive top-up fees and indicate the amount of such fees.

B Additional Tables

Table B.1: Effects of the Targeted Voucher Program on School Entry and Exit

	<i>Entry</i>			<i>Exit</i>		
	public	private- voucher	private- non-voucher	public	private- voucher	private- non-voucher
	(1)	(2)	(3)	(4)	(5)	(6)
-3	-0.072 (0.151)	0.079 (0.224)	0.148 (0.126)	-0.225 (0.196)	0.131 (0.182)	-0.291 (0.139)
-2	0.025 (0.086)	-0.073 (0.215)	0.072 (0.086)	-0.093 (0.268)	0.150 (0.156)	-0.034 (0.094)
0	-0.010 (0.069)	-0.041 (0.204)	0.093 (0.158)	-0.210 (0.173)	0.194 (0.165)	-0.005 (0.113)
1	-0.082 (0.083)	-0.105 (0.186)	0.204 (0.146)	-0.076 (0.187)	0.322 (0.179)	0.073 (0.102)
2	-0.013 (0.071)	0.330 (0.158)	0.258 (0.142)	0.111 (0.220)	0.352 (0.167)	0.146 (0.094)
3	-0.206 (0.106)	0.454 (0.159)	0.278 (0.116)	0.090 (0.239)	0.373 (0.174)	0.027 (0.126)
4	-0.088 (0.109)	-0.067 (0.185)	0.225 (0.140)	-0.088 (0.184)	0.227 (0.161)	0.089 (0.080)
5	0.008 (0.085)	0.309 (0.183)	0.203 (0.137)	-0.343 (0.205)	0.271 (0.195)	0.035 (0.070)
6	-0.187 (0.101)	0.675 (0.164)	0.170 (0.127)	0.271 (0.198)	0.304 (0.151)	0.084 (0.106)
7	0.041 (0.077)	0.574 (0.153)	0.226 (0.142)	0.013 (0.180)	0.137 (0.154)	0.096 (0.103)
observations	3,795	3,795	3,795	3,795	3,795	3,795
R^2	0.167	0.412	0.281	0.229	0.372	0.348

Notes: A school enters the market in year t if it is observed in period t , but is not in the market in year $t - 1$. A school exits the market in year t if it is observed in year $t - 1$, but not in t . Results come from the estimation of event study regressions that use school data from three years prior to the introduction of the 2008 targeted voucher program, and from the following eight years. Clustered (at the municipality level) standard errors are in parentheses.

Table B.2: Effects of the Targeted Voucher Program on School Share of Multigrade Classes

	public	private- voucher	private- non-voucher
	(1)	(2)	(3)
-4	-0.003 (0.032)	0.015 (0.030)	0.027 (0.054)
-3	-0.025 (0.036)	-0.002 (0.021)	-0.017 (0.037)
-2	0.003 (0.028)	0.001 (0.012)	0.030 (0.059)
0	0.003 (0.012)	0.015 (0.012)	-0.065 (0.034)
1	0.023 (0.020)	-0.012 (0.024)	-0.091 (0.042)
2	-0.047 (0.032)	-0.038 (0.028)	-0.096 (0.047)
3	-0.190 (0.041)	-0.302 (0.041)	-0.106 (0.053)
4	-0.103 (0.039)	-0.299 (0.036)	-0.070 (0.060)
5	-0.367 (0.057)	-0.621 (0.045)	-0.083 (0.046)
6	-0.394 (0.052)	-0.557 (0.041)	-0.073 (0.048)
7	-0.370 (0.052)	-0.485 (0.043)	-0.055 (0.043)
observations	59,566	38,271	5,110
R^2	0.309	0.526	0.147

Notes: Results come from the estimation of event study regressions that use school data from four years prior to the introduction of the 2008 targeted voucher program, and from the following eight years. Clustered (at the municipality level) standard errors are in parentheses.

Table B.3: Effects of the Targeted Voucher Program on School Class Size

	public	private- voucher	private- non-voucher
	(1)	(2)	(3)
-4	0.900 (0.586)	1.044 (0.696)	0.652 (2.065)
-3	0.373 (0.535)	0.921 (0.575)	-1.672 (2.190)
-2	0.482 (0.330)	0.488 (0.381)	0.539 (1.945)
0	0.288 (0.248)	-0.477 (0.375)	2.135 (1.356)
1	0.114 (0.387)	-1.355 (0.562)	7.145 (3.553)
2	0.350 (0.452)	-1.922 (0.684)	2.546 (1.975)
3	-0.705 (0.509)	-4.641 (0.790)	1.617 (2.523)
4	0.491 (0.522)	-5.287 (0.685)	0.763 (2.451)
5	-1.901 (0.592)	-9.243 (0.818)	2.473 (2.584)
6	-1.253 (0.609)	-8.711 (0.813)	3.268 (2.417)
7	-0.726 (0.646)	-7.986 (0.847)	4.939 (2.751)
observations	59,566	38,271	5,110
R^2	0.457	0.367	0.173

Notes: Results come from the estimation of event study regressions that use school data from four years prior to the introduction of the 2008 targeted voucher program, and from the following eight years. Clustered (at the municipality level) standard errors are in parentheses.

Table B.4: Effects of the Targeted Voucher Program on School Pupil-teacher Ratio

	public (1)	private- voucher (2)	private- non-voucher (3)
-4	-0.574 (1.249)	-0.788 (0.823)	-0.858 (2.455)
-3	0.040 (0.652)	0.188 (0.729)	-1.239 (1.506)
-2	-0.432 (0.470)	-0.314 (0.598)	0.875 (1.190)
0	-0.319 (0.388)	-1.265 (0.357)	1.400 (1.073)
1	-3.320 (1.831)	-0.750 (1.036)	-0.200 (1.608)
2	-1.521 (0.999)	-1.443 (0.679)	-0.137 (1.482)
3	0.775 (0.750)	-1.173 (0.592)	-10.489 (11.766)
4	1.746 (0.631)	-3.123 (0.821)	5.813 (3.299)
5	2.265 (0.608)	-1.938 (0.690)	4.080 (2.464)
6	3.221 (0.614)	-2.423 (0.674)	13.207 (10.054)
7	3.777 (0.632)	-2.737 (0.748)	-16.837 (17.091)
observations	59,566	38,271	5,110
R^2	0.203	0.191	0.056

Notes: Results come from the estimation of event study regressions that use school data from four years prior to the introduction of the 2008 targeted voucher program, and from the following eight years. Clustered (at the municipality level) standard errors are in parentheses.

Table B.5: Effects of the Targeted Voucher Program on School Share of Indefinite Contracts

	public	private- voucher	private- non-voucher
	(1)	(2)	(3)
-4	0.014 (0.024)	-0.081 (0.030)	0.032 (0.101)
-3	0.024 (0.015)	-0.019 (0.021)	0.115 (0.069)
-2	-0.008 (0.010)	-0.026 (0.018)	0.067 (0.060)
0	0.021 (0.016)	-0.010 (0.015)	-0.087 (0.075)
1	-0.010 (0.023)	-0.010 (0.022)	-0.053 (0.084)
2	-0.017 (0.026)	-0.021 (0.021)	-0.103 (0.089)
3	0.043 (0.035)	-0.024 (0.025)	0.022 (0.089)
4	0.046 (0.036)	0.022 (0.028)	-0.060 (0.107)
5	0.033 (0.038)	0.041 (0.036)	-0.064 (0.108)
6	0.056 (0.040)	0.015 (0.039)	-0.106 (0.106)
7	0.020 (0.041)	0.027 (0.040)	-0.011 (0.108)
observations	59,566	38,271	5,110
R^2	0.326	0.158	0.174

Notes: Results come from the estimation of event study regressions that use school data from four years prior to the introduction of the 2008 targeted voucher program, and from the following eight years. Clustered (at the municipality level) standard errors are in parentheses.

Table B.6: Effects of the Targeted Voucher Program on School Share of Specialized Teachers

	public	private- voucher	private- non-voucher
	(1)	(2)	(3)
-4	0.005 (0.007)	0.018 (0.014)	-0.199 (0.060)
-3	0.001 (0.007)	0.004 (0.013)	-0.058 (0.054)
-2	-0.001 (0.004)	-0.001 (0.008)	-0.048 (0.026)
0	-0.005 (0.007)	-0.014 (0.007)	-0.005 (0.035)
1	-0.001 (0.010)	-0.041 (0.009)	-0.014 (0.044)
2	-0.007 (0.011)	-0.046 (0.009)	0.012 (0.055)
3	-0.135 (0.020)	-0.051 (0.018)	-0.138 (0.093)
4	-0.140 (0.020)	-0.069 (0.020)	-0.090 (0.065)
5	-0.124 (0.029)	-0.058 (0.031)	-0.080 (0.093)
6	-0.123 (0.029)	-0.044 (0.032)	-0.127 (0.110)
7	-0.088 (0.031)	-0.021 (0.037)	-0.145 (0.110)
observations	59,566	38,271	5,110
R^2	0.506	0.459	0.335

Notes: A specialized teacher is one with extra training in one or more subjects. Results come from the estimation of event study regressions that use school data from four years prior to the introduction of the 2008 targeted voucher program, and from the following eight years. Clustered (at the municipality level) standard errors are in parentheses.

Table B.7: Effects of the Targeted Voucher Program on School Weekly Teaching Hours per Teacher

	public (1)	private- voucher (2)	private- non-voucher (3)
-4	1.285 (0.503)	-0.340 (0.383)	2.257 (1.808)
-3	0.472 (0.345)	-0.103 (0.375)	1.419 (1.294)
-2	-0.158 (0.214)	-0.442 (0.371)	1.365 (0.887)
0	0.589 (0.453)	-0.483 (0.300)	0.074 (0.677)
1	0.665 (0.481)	-0.490 (0.384)	-0.993 (1.168)
2	1.323 (0.569)	-1.167 (0.423)	-1.646 (1.233)
3	0.428 (0.758)	-3.163 (0.606)	-0.066 (2.170)
4	-0.232 (0.727)	-3.657 (0.596)	-0.200 (1.876)
5	-1.226 (0.816)	-3.745 (0.601)	-1.497 (1.965)
6	-2.417 (0.806)	-4.695 (0.678)	-1.687 (1.919)
7	-3.157 (0.929)	-5.042 (0.728)	-0.896 (2.506)
observations	59,566	38,271	5,110
R^2	0.308	0.292	0.163

Notes: Results come from the estimation of event study regressions that use school data from four years prior to the introduction of the 2008 targeted voucher program, and from the following eight years. Clustered (at the municipality level) standard errors are in parentheses.

Table B.8: Effects of the Targeted Voucher Program on Private-voucher School Top-up Fees

	low income	higher income
	(1)	(2)
-4	0.941 (1.700)	1.148 (1.724)
-3	-1.040 (1.773)	-1.039 (1.776)
-2	-5.368 (2.269)	-5.384 (2.270)
0	5.071 (1.693)	-2.352 (1.630)
1	2.605 (2.056)	-7.909 (2.104)
2	3.216 (1.796)	-7.815 (1.866)
3	5.808 (2.077)	-5.871 (1.987)
4	4.876 (2.100)	-9.798 (1.989)
5	2.493 (2.429)	-11.158 (2.434)
6	5.020 (2.505)	-8.227 (2.634)
7	4.722 (2.336)	-8.734 (2.480)
observations	38,271	38,271
R^2	0.272	0.218

Notes: Fee values are real, and are converted from Chilean pesos to US dollars according to the exchange rate of Ch\$686.52 per US dollar, as of May 16, 2016. Results come from the estimation of event study regressions that use school data from four years prior to the introduction of the 2008 targeted voucher program, and from the following eight years. Column (1) shows the effects of the reform on the top-up fees faced by low income students that are eligible to participate in the program, and column (2) shows the effects of the reform on the top-up fees faced by higher income students. Clustered (at the municipality level) standard errors are in parentheses.